

Hands-on Lab: Advanced Dashboard Capabilities in Cognos Analytics

Purpose of the Lab:

This lab focuses on enhancing skills in utilizing advanced features of IBM Cognos Analytics to create more dynamic and interactive dashboards. It delves into creating calculations, manipulating data points within visualizations, applying top/bottom settings on visualizations, and constructing navigation paths. Additionally, the lab provides hands-on experience in filtering data within a dashboard. The exercises are designed to provide a deeper understanding of how to leverage Cognos Analytics for more complex data analysis and visualization tasks, moving beyond basic dashboard creation.

Benefits of Learning the Lab:

Engaging in this lab offers several key benefits for those interested in data analytics and visualization. You will acquire practical skills in advanced dashboarding techniques, such as creating custom calculations, effectively filtering and manipulating data, and utilizing Cognos Analytics to its full potential for comprehensive data analysis. These skills are vital for professionals in data analysis, business intelligence, and decision-making roles, as they allow for more nuanced and insightful analysis of data. The ability to create interactive and detailed dashboards enhances one's capability to present data in a more engaging and informative manner. This knowledge is particularly beneficial for those seeking to improve their data presentation skills, making complex data more accessible and actionable for decision-makers. Overall, the lab provides a strong foundation in advanced data visualization techniques, making it a valuable learning experience for advancing one's career in the field of data analytics.

Software Used in this Lab

Like the videos in the course, for the hands-on labs, we will be using IBM Cognos Analytics trial version (currently limited to 30 days), as this is available at no charge.

Dataset Used in this Lab

The dataset used in this lab comes from the VM designed to showcase IBM Cognos Analytics. This dataset is published by IBM. You can download the dataset file directly from here: [CustomerLoyaltyProgram.csv](#) 

Objectives

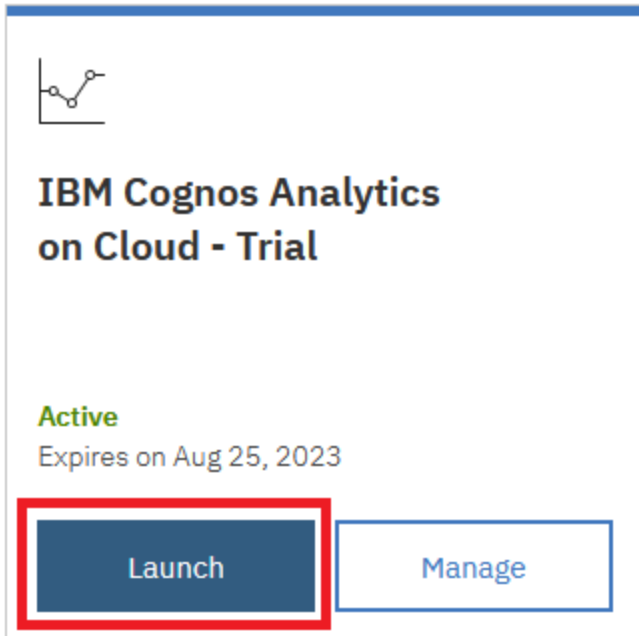
After completing this lab, you will be able to:

- Start a new dashboard
- Create calculations
- Keep/exclude data points from a visualization
- Set top/bottom on a visualization
- Create and leverage navigation paths
- Filter data in a dashboard

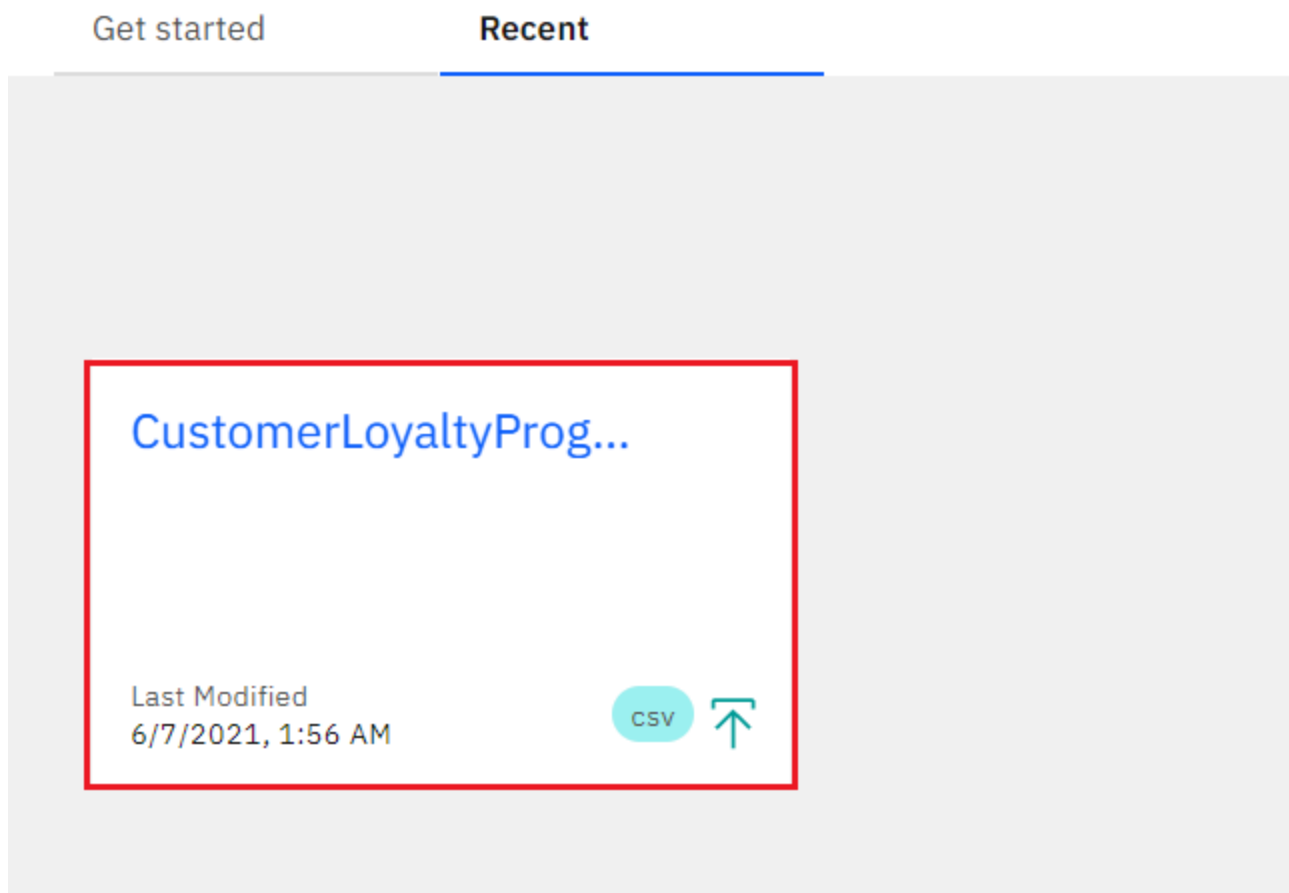
Exercise 1: Start a New Dashboard

In this exercise, you will start a new dashboard for working with advanced Cognos Analytics dashboard capabilities.

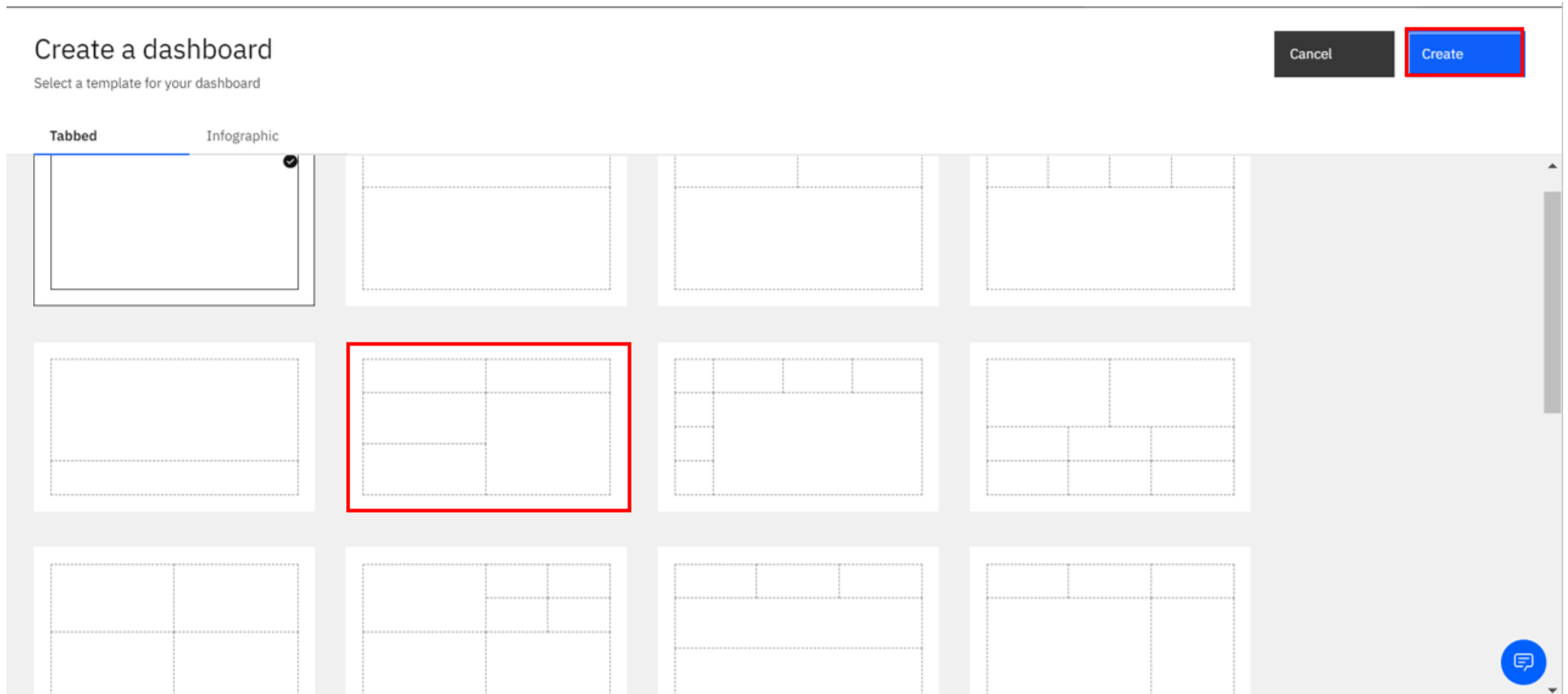
1. To sign in to the Cognos Analytics platform with your IBMid, go to myibm.ibm.com/dashboard/ .
2. Enter your IBMid and password.
3. Scroll down and click **Launch**.



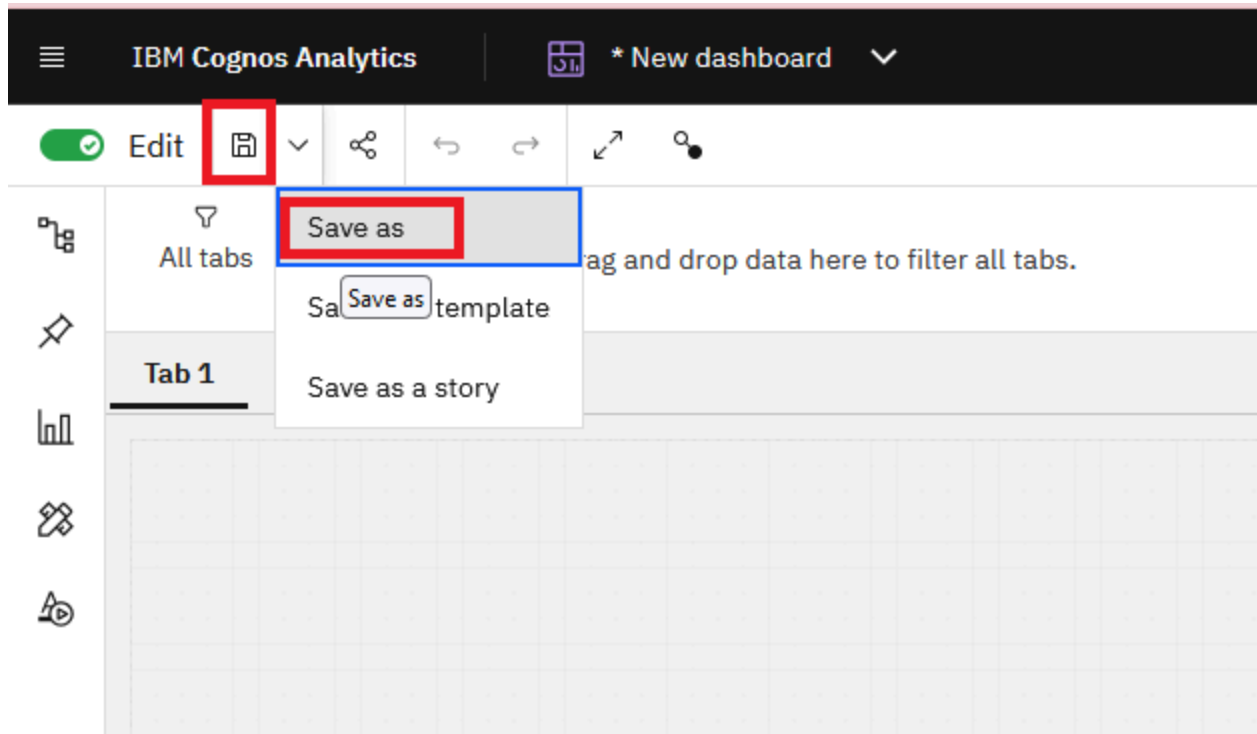
4. From the **Recent** section, click the uploaded data file **CustomerLoyaltyProgram.csv**.



5. The template window will be displayed, allowing you to select the type of dashboard and the template style. Select the **Tabbed** dashboard style. This will allow you to have multiple pages for your dashboards. Select the *five-panel template*, then click **Create**.



6. To save the newly created dashboard, press **CTRL+S** or click the **Save** icon and then click **Save as**.



7. A new **Save as** window will pop up. Follow the steps as displayed below to save your dashboard as **Advanced Dashboard** in the **My content** section.

 * New dashboard 

29

Save as 

Name

Advanced Dashboard

Step 1 - Enter the name for the dashboard

Selected destination: My content

My content

Team content

Step 2 - Select My content

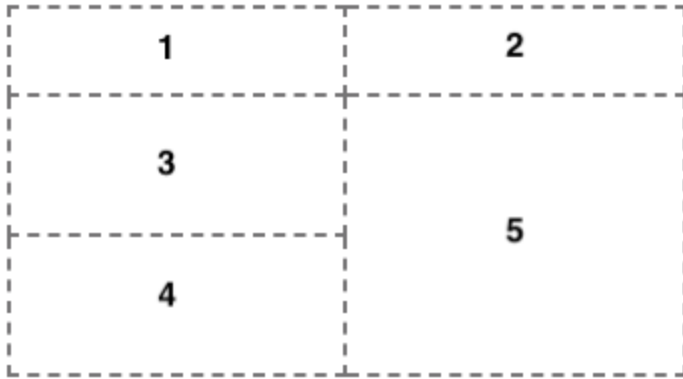
Name	Type	Last Accessed
 CLP_Dashboard	Dashboard	7/27/2023, 3:35 AM
 CustomerLoyaltyProgram.csv 	Uploaded file	7/26/2023, 5:22 AM

Cancel

Save

Step 3 - Click on Save

8. As you build the dashboard, the location placement for visualization widgets in the dashboard template will be referenced using the following Panel numbers.



9. From the **Navigation** panel, select **Sources** to ensure the data source panel is open in the left pane.
10. From the data source panel, select **Revenue** and drag it to the center of **Panel 1**, releasing it once you see the drop zone turn blue.

The screenshot shows the IBM Cognos Analytics Advanced Dashboard interface. The top navigation bar includes the IBM Cognos Analytics logo, the 'Advanced Dashboard' tab, and a user profile icon. The main toolbar contains icons for Edit, Save, Share, Undo, Redo, and a search icon. The right sidebar has tabs for Analytics, Filters, Fields, and Properties. The left sidebar shows a list of data sources under 'Selected sources /', with 'CustomerLoyaltyProgram.csv' selected. Below this is a search bar and a list of fields: Quarter, MonthsAsMember, LoyaltyStatus, Product Line, Coupon Response, Count, Quantity Sold, Unit Sale Price, Unit Cost, Revenue (highlighted), Customer ...ime Value, and Loyalty Count. The main workspace displays a grid for 'Tab 1'. A blue box labeled 'Revenue' is being dragged onto the grid, with a 'maximize' button visible next to it. The grid is divided into sections by dashed lines.

11. Click the summary chart in Panel 1 to bring it into focus. From the on-demand toolbar that appears in the main toolbar, click **Summarize**, and then select **Average**.

The screenshot shows the IBM Cognos Analytics interface. The top navigation bar includes the IBM Cognos Analytics logo and a dropdown menu for "Advanced Dashboard". Below the navigation bar is a toolbar with various icons, including a red box highlighting the "Summary" icon (a square with a sigma symbol). The main workspace is divided into two panels. The left panel, titled "Selected sources /", shows a list of data sources for "CustomerLoyaltyProgram.csv". The right panel, titled "Tab 1", shows a summary chart configuration. A red box highlights the "Average" option in the "Summary" dropdown menu. The chart area shows a grid with a red line representing the "Revenue" data source. The "Summary" dropdown menu is open, showing options: Average, Sum, Minimum, Maximum, Count, Count distinct, and Auto (Sum) (which is selected with a checkmark).

12. In the summary chart in Panel 1, select the title of the visualization and change it to *Average Revenue*.

13. From the **Navigation** panel, select **Widgets** to open the widgets panel. Drag and drop **Money coin** from **Shapes** to the center of Panel 1.

The screenshot shows the IBM Cognos Analytics Advanced Dashboard editor interface. The top bar displays 'IBM Cognos Analytics' and '* Advanced Dashboard'. The left sidebar contains a grid of icons for various dashboard elements, with the 'Average Revenue' icon (a blue circle with a white dollar sign) highlighted by a red square. The main workspace shows a visualization titled 'Tab 1' with the label 'Average Revenue' and a large value of '2.71K'. A red arrow points to the 'Average Revenue' icon in the sidebar, indicating it should be added to the dashboard. The visualization area also shows a 'Revenue' label and a blue circle with a white dollar sign, which is the icon for the 'Average Revenue' measure.

14. To save the current work in the dashboard, press **CTRL+S** or click **Save** in the toolbar.

15. Your Panel 1 visualization should look similar to the one below:

Average Revenue



 = 100 Revenue

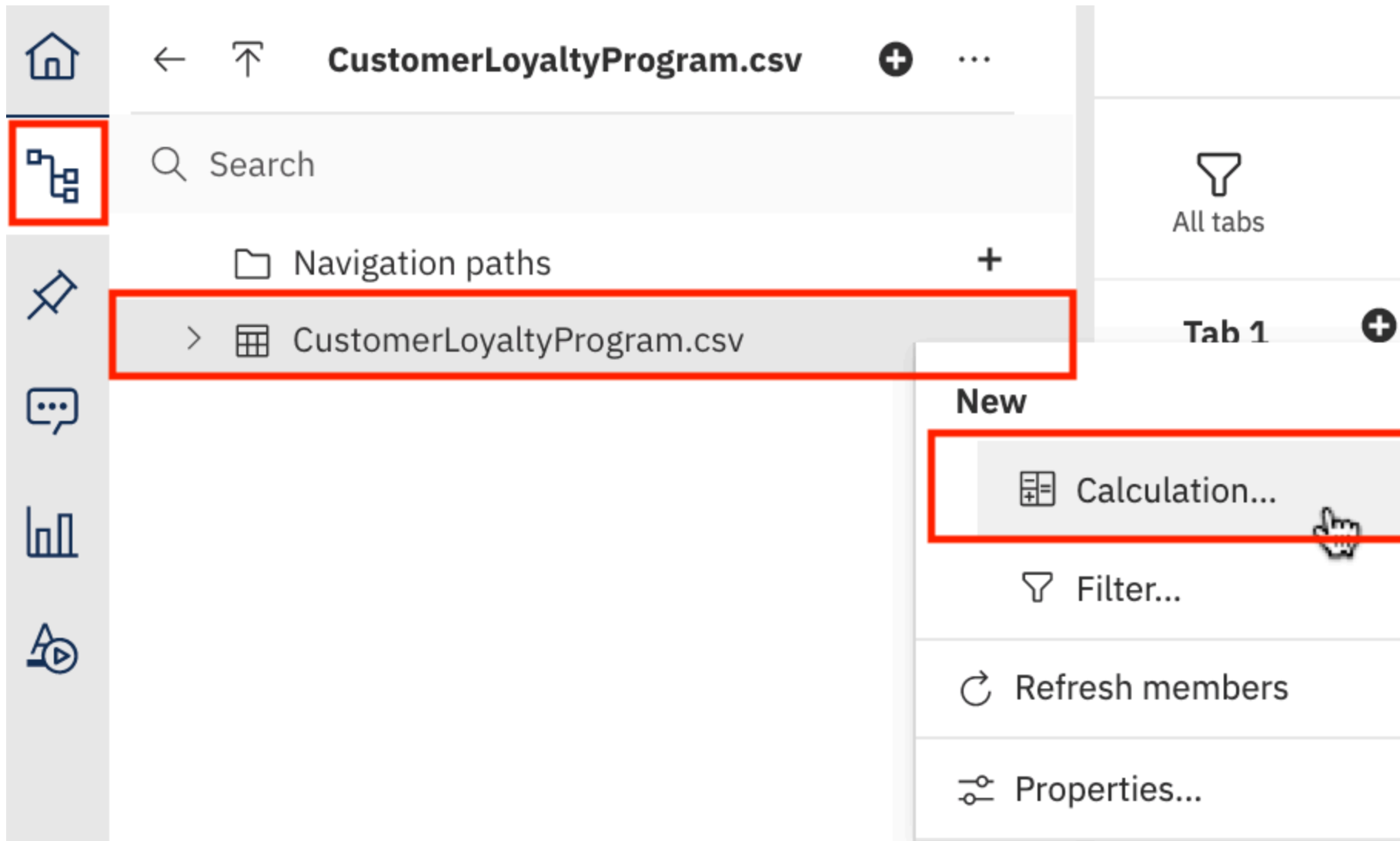
Exercise 2: Working with Advanced Cognos Analytics dashboard capabilities

In this exercise, you will practice some advanced Cognos Analytics dashboard capabilities.

- Task A: Create calculations
- Task B: Keep/Exclude Data Points from a visualization
- Task C: Set Top/Bottom on a visualization
- Task D: Create and Leverage navigation paths
- Task E: Filter Data in the current tab

Task A: Create Calculations

1. From the **Navigation** panel, select **Sources** to open the data source panel if it is not already open. The data source panel displays the uploaded file **CustomerLoyaltyProgram.csv** as the selected source.
2. Right-click the **CustomerLoyaltyProgram.csv** data source and select **Calculation**.



3. Change the calculation name to **Margin**. From the **Components** panel, drag **Unit Sale Price** to the **Expression** field, type a space, then the minus sign, -, to the right of it, and then drag **Unit Cost** to the right of that. Click **OK**.

IBM Cognos Analytics Advanced Dashboard

Create calculation

Name: **Margin**

Expression: **1 Unit Sale Price**

Components:

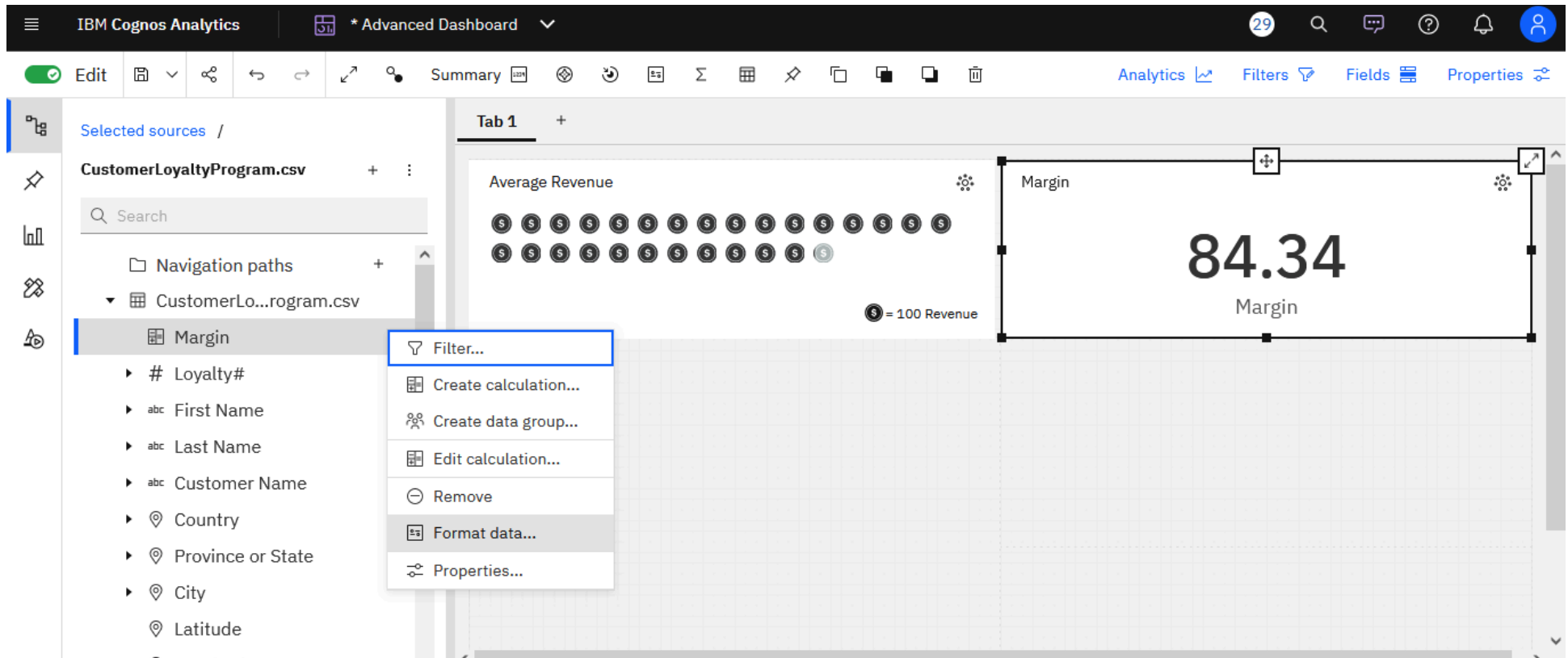
- MonthsAsMember
- abc LoyaltyStatus
- abc Product Line
- abc Coupo...ponse
- Count
- Quantity Sold
- Unit Sale Price**
- Unit Cost
- Revenue
- Custom...e Value
- Loyalty Count

Information

Expression

1 Unit_Sale_Price - Unit_Cost

4. In the data source panel, expand CustomerLoyaltyProgram.csv if needed, and drag **Margin** to the center of **Panel 2**, releasing it once you see the drop zone turn blue.
5. Right-click the margin chart in Panel 2, point to **Summarize**, and then select **Average**.
6. From the data source panel, right-click on **Margin** and click **Format data**.



7. In the **Format type** list, select **Currency**.

Advanced Dashboard

Data format

Column: Margin

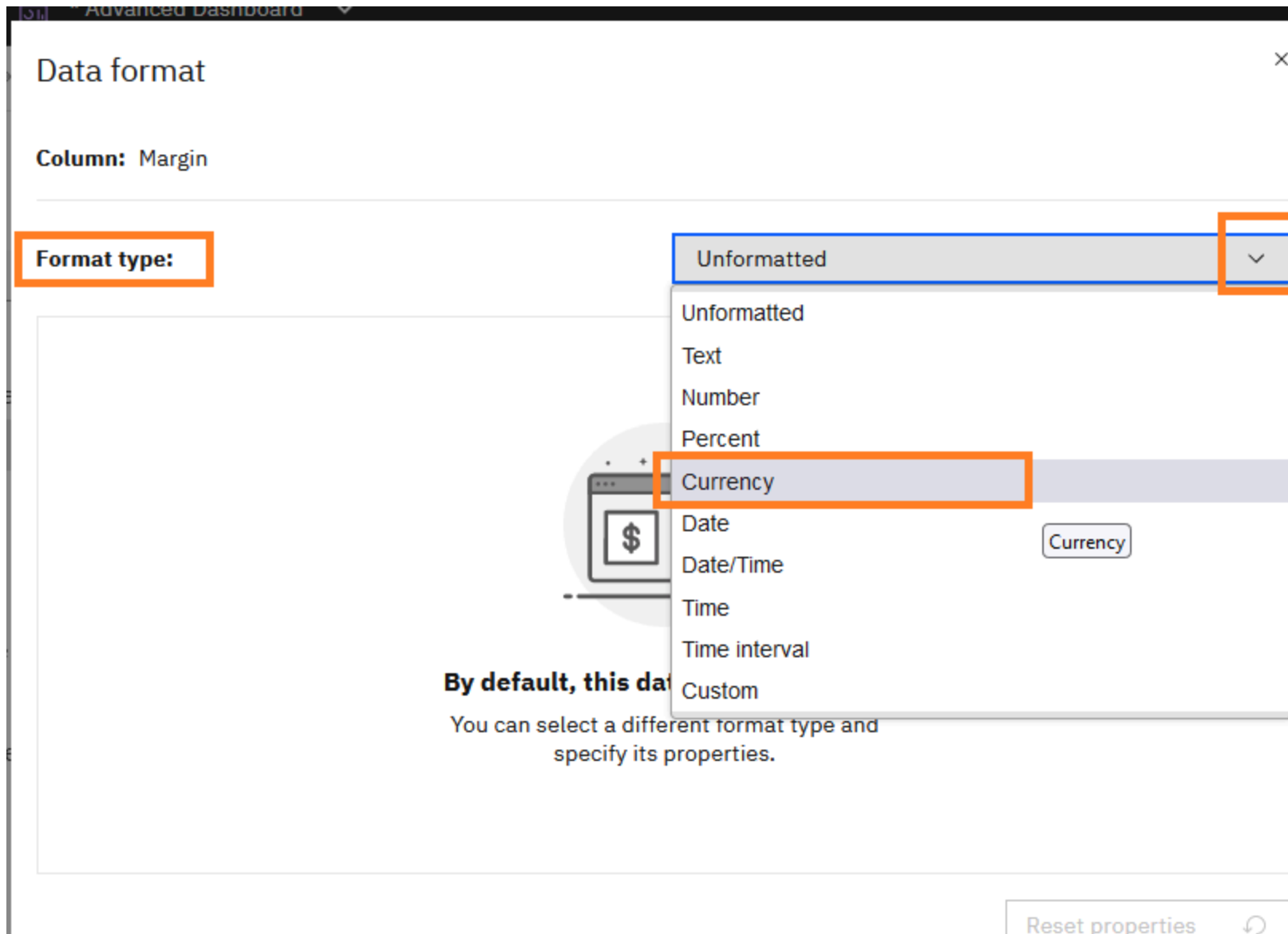
Format type:

Unformatted

Unformatted
Text
Number
Percent
Currency
Date
Date/Time
Time
Time interval
Custom

By default, this data is formatted as currency. You can select a different format type and specify its properties.

Reset properties



8. Select **\$ (USD) - United States of America, dollar** as the currency and click **OK** at the bottom.

Data format



Column: Margin

Format type: •

Currency



Currency

Currency symbol

Number of decimal places

Use thousands separator

Currency display

Missing value characters

Default



lei (RON) - Romania, lei

руб. (RUB) - Russia, ruble

ر.س. (SAR) - Saudi Arabia, riyal

kr (SEK) - Sweden, krona

S\$ (SGD) - Singapore, dollar

฿ (THB) - Thailand, baht

TL (TRY) - Turkey, lira

NT\$ (TWD) - Taiwan, dollar

₴ (UAH) - Ukraine, hryvnia

\$ (USD) - United States of America, dollar

Bs.F. (VEF) - Venezuela, bolívar fuerte

₫ (VND) - Việt Nam, đồng

R (ZAR) - South Africa, rand

\$ (USD) - United States of America, dollar

9. In the margin chart in Panel 2, select the title of the visualization and change it to *Average Margin*.

10. To save the current work in the dashboard, press **CTRL+S** or click **Save** in the main toolbar.

11. Your Panel 2 visualization should look similar to the one below:

Average Margin

\$84.36
Margin

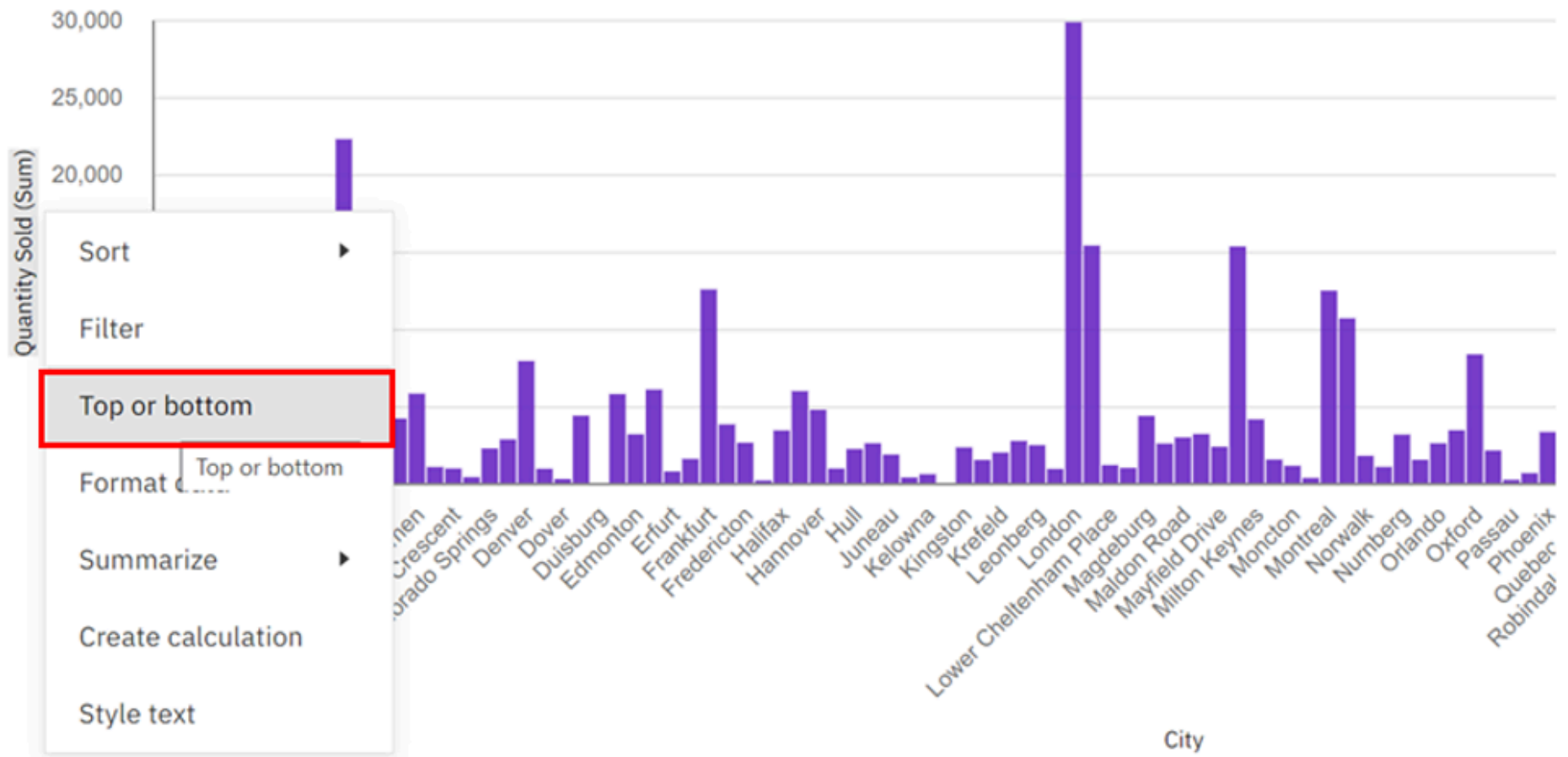
Task B: Keep/Exclude Data Points from a Visualization

1. In the data source panel, expand CustomerLoyaltyProgram.csv if needed. Press the **CTRL** key and select **Revenue** and **Product Line** and drag them both to the center of **Panel 3**, releasing them once you see the drop zone turn blue.
2. From the data source panel, drag **Location Code** to the **Color** drop zone of **Panel 3**.
3. Right-click the **Suburban** data point in the Panel 3 visualization, and select **Exclude**.
4. To save the current work in the dashboard, press **CTRL+S** or click **Save** in the main toolbar.
5. Your Panel 3 visualization should look similar to the one below:

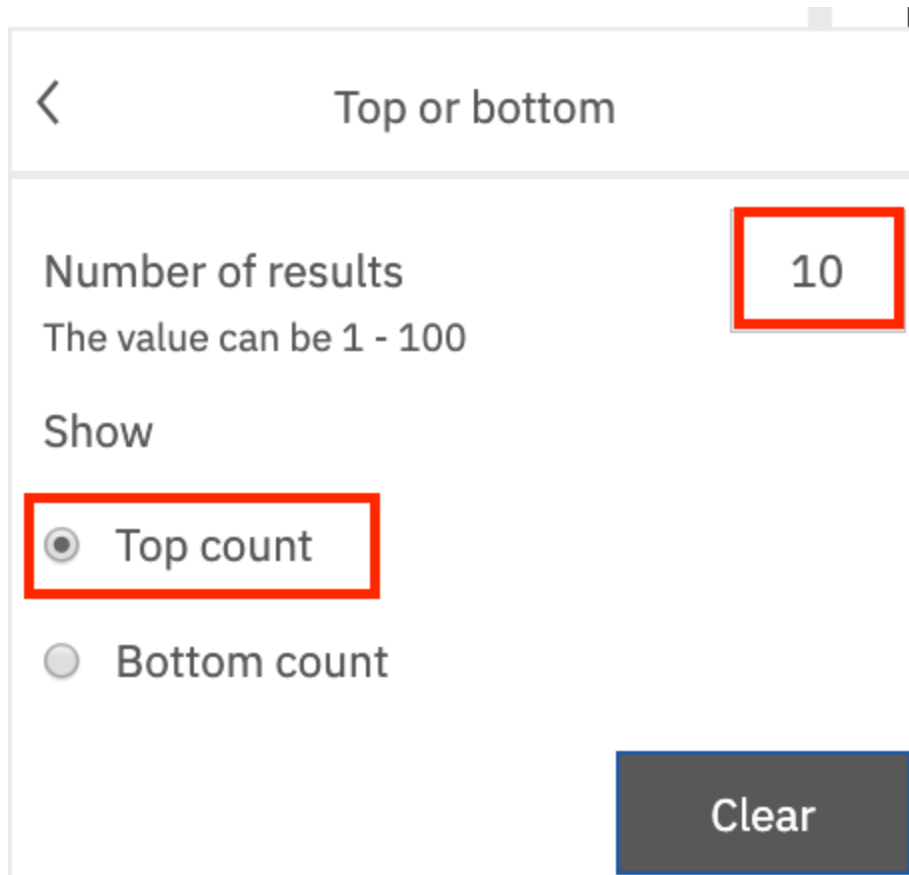
Task C: Set Top/Bottom on a Visualization

1. From the data source panel, press the **CTRL** key and select **Quantity Sold** and **City**, and drag them both to the center of **Panel 4**, releasing them once you see the drop zone turn blue.
2. Click the chart in Panel 4 to bring it into focus and render the on-demand toolbar.
3. Click the **Change visualization** button in the on-demand toolbar (which will currently say **Map**), then expand **All visualizations**, if needed, and select **Column**.
4. In Panel 4, right-click the axis label **Quantity Sold (Sum)** down the left side of the chart and select **Top or bottom**.

Quantity Sold by City



5. Ensure the value of **Number of results** is set to **10**, then select **Top count**.



< Top or bottom

Number of results
The value can be 1 - 100

10

Show

☒ Top count

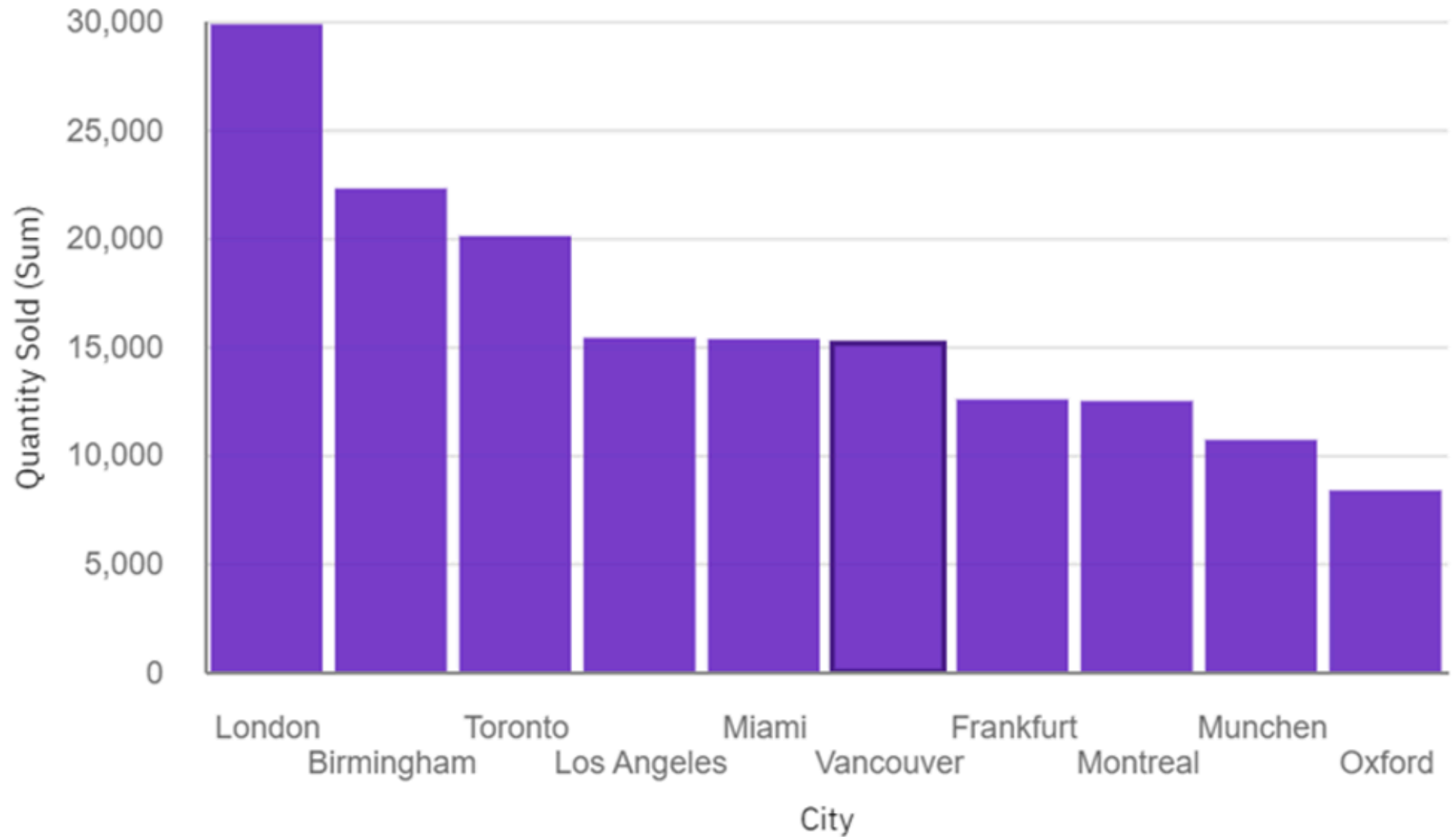
☐ Bottom count

Clear

6. In the column chart in Panel 4, select the title of the visualization and change it to *Top 10 Quantity Sold by City*.
7. To save the current work in the dashboard, press **CTRL+S** or click **Save** in the main toolbar.
8. Your Panel 4 visualization should look similar to the one below:

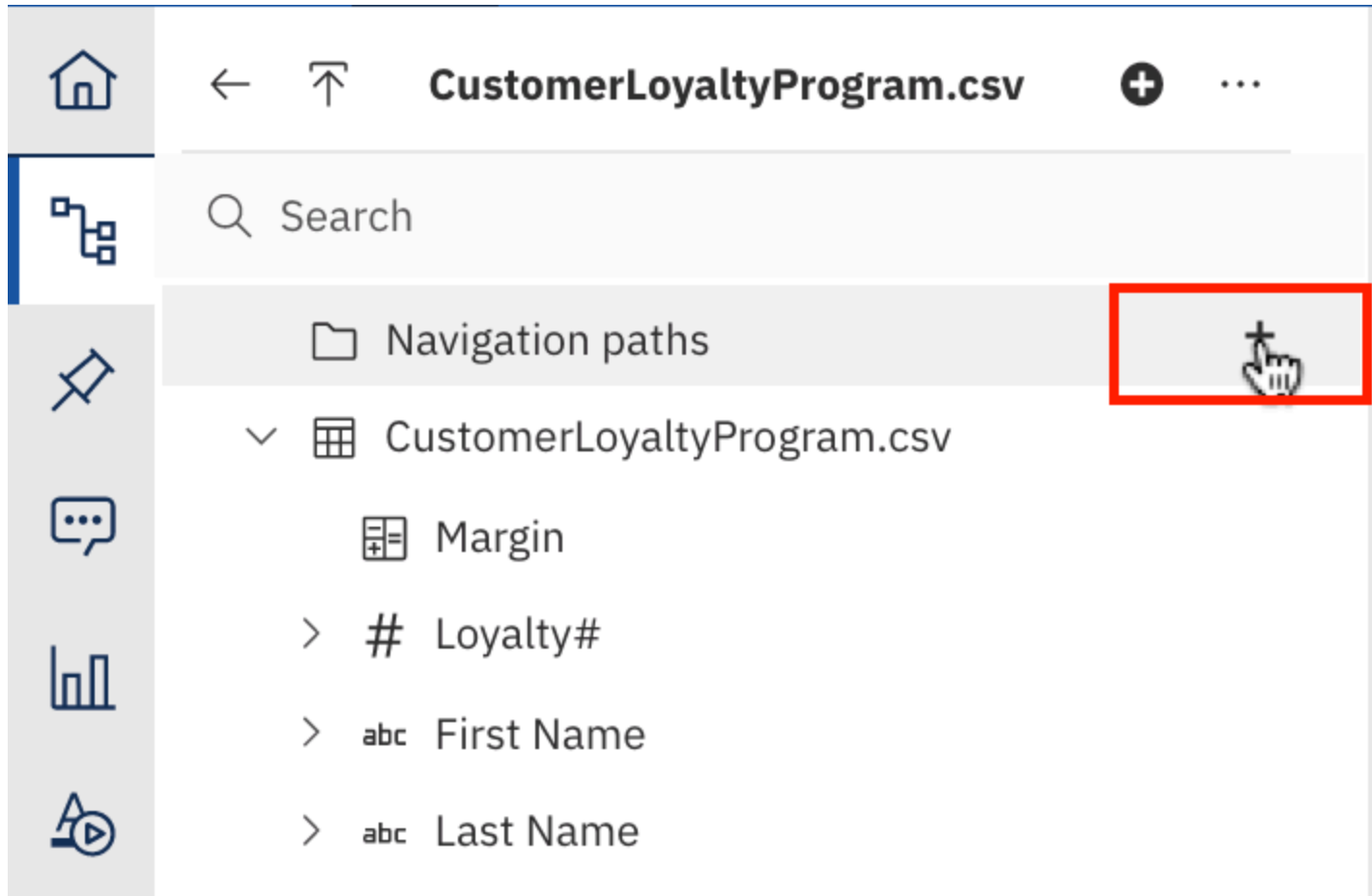


Quantity Sold by City



Task D: Create and leverage navigation paths

1. In the data source panel on the left, scroll to the top of the list and click the **plus sign** labeled **Create navigation path** to the right of **Navigation paths**.



2. In the **Create navigation path** dialog box, expand **CustomerLoyaltyProgram.csv**, if needed. Drag **Order Year**, **Quarter**, **Country**, and **City** sequentially to the right hand panel of the dialog box, maintaining the order (shown in the image below). Once done, click **OK**.

Create navigation path



Search

CustomerLoyaltyProgram.csv

- # Loyalty#
- abc First Name
- abc Last Name
- abc Customer Name
- 📍 Country
- 📍 Province or State
- 📍 City
- 📍 Latitude
- 📍 Longitude
- 📍 Postal code
- abc Gender
- abc Education
- abc Location Code

Name Order Year - City

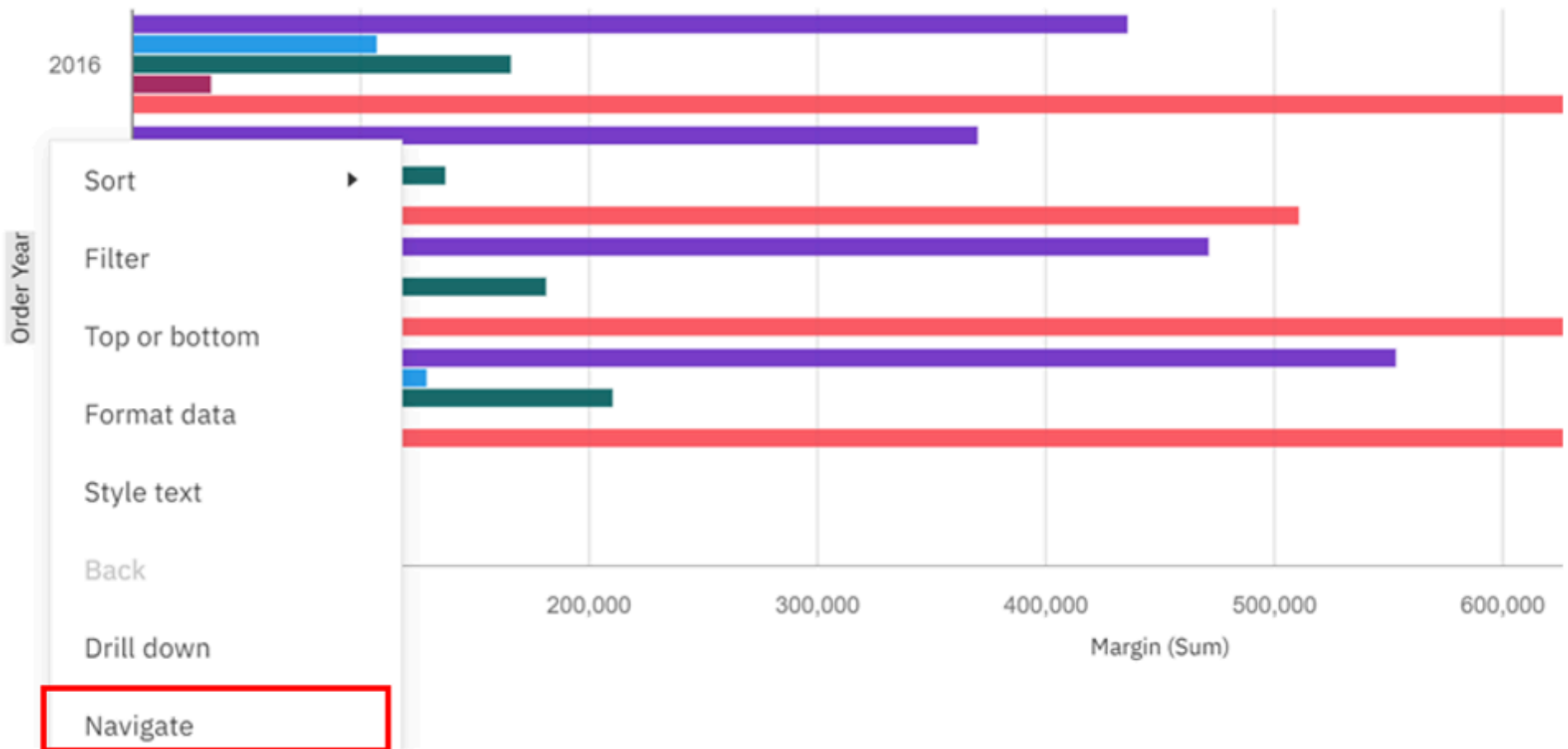
Select and order the columns to use in the navigation path.

- 🕒 Order Year
CustomerLoyaltyProgram.csv
- 🕒 Quarter
CustomerLoyaltyProgram.csv
- 📍 Country
CustomerLoyaltyProgram.csv
- 📍 City
CustomerLoyaltyProgram.csv

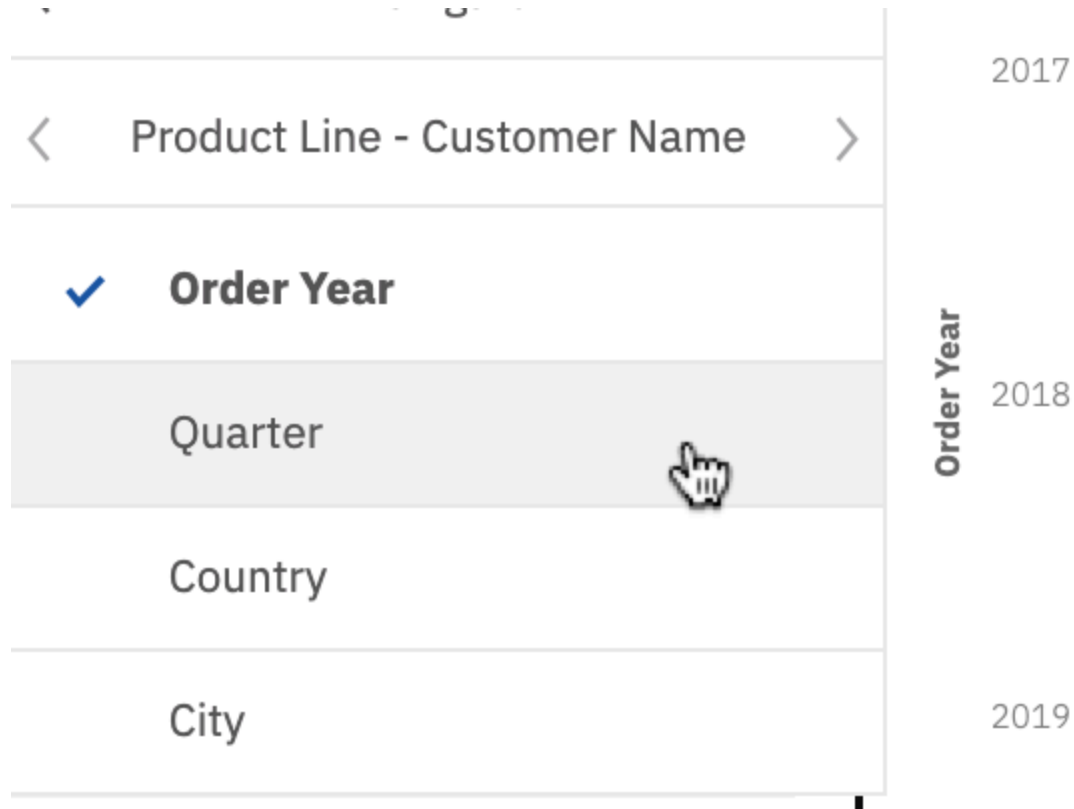
OK

Cancel

3. From the data source panel, press the **CTRL** key and select **Margin**, **Product Line**, and **Order Year** and drag them to the center of **Panel 5**, releasing them once you see the drop zone turn blue.
4. Click the line chart in Panel 5 to bring it into focus and render the on-demand toolbar.
5. Click the **Change visualization** button in the on-demand toolbar (which will currently say **Line**), then expand **All visualizations**, if needed, and select **Bar**.
6. In Panel 5, right-click the axis label **Order Year** down the left side of the chart, and select **Navigate**.

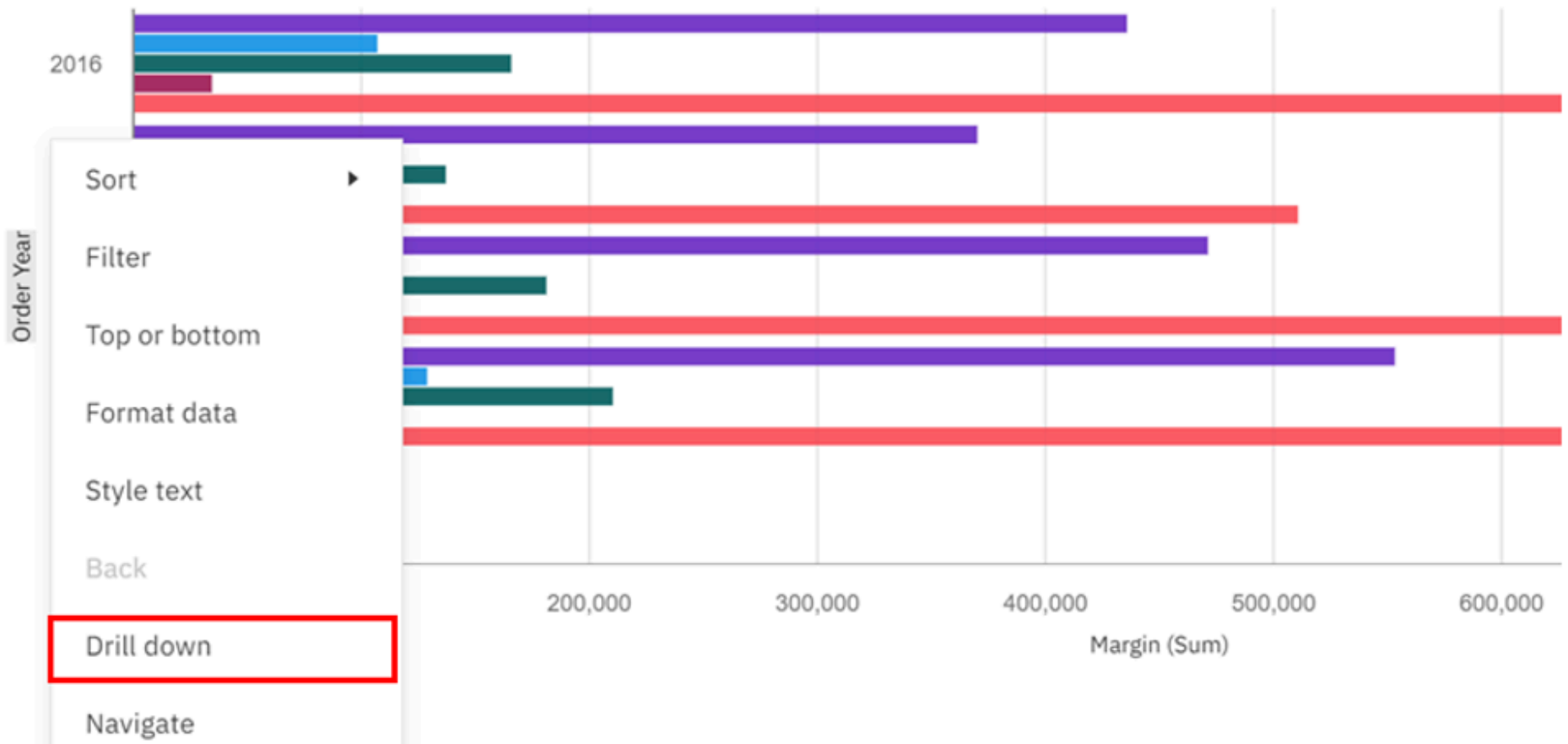


7. One by one, select the **Order Year**, **Quarter**, **Country**, and **City** options in the **Navigate** dialog box to view the different navigation paths and observe the resulting visualization in Panel 5 as you select each one. Lastly, keep the **Order Year** option selected.



8. Alternative interactive way with Drill down/back:

- In the bar chart in Panel 5, right-click the **2016 - Smart Electronics** bar of the bar chart, and select **Drill down**.



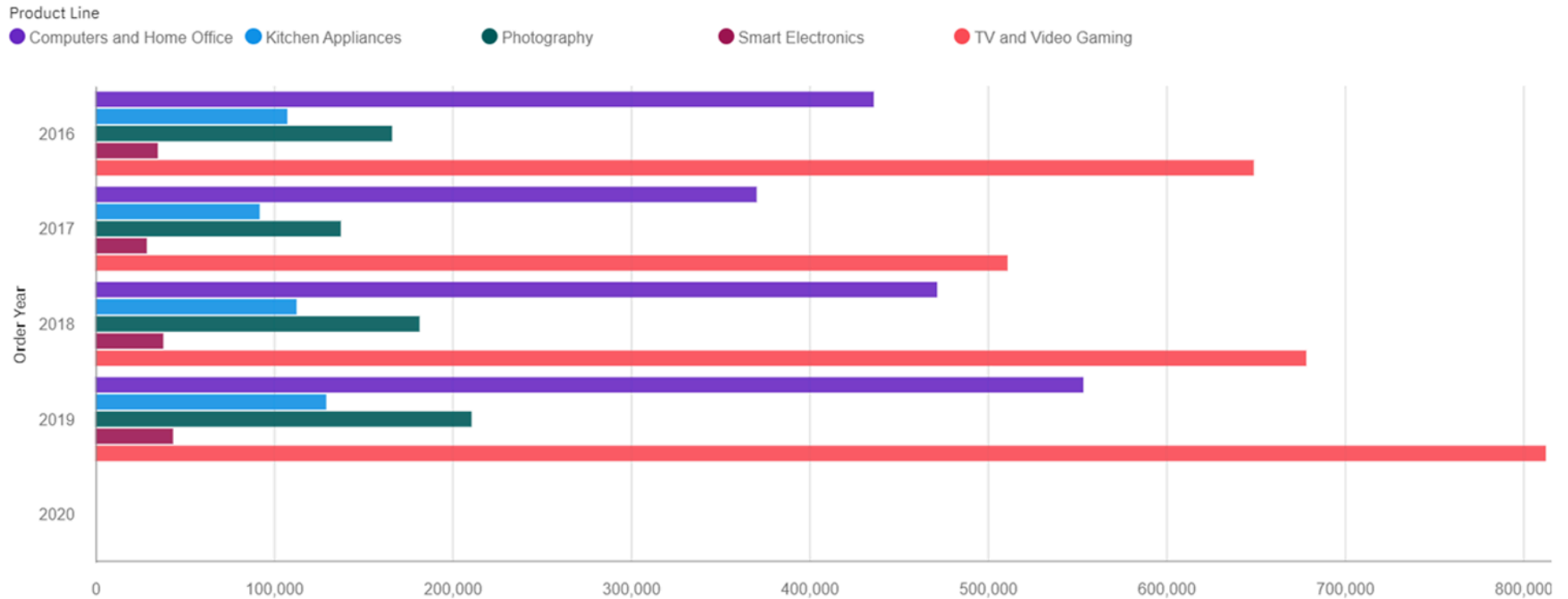
- Now right-click the **Q1 - Smart Electronics** bar of the bar chart, and select **Back**.



10. To save the current work in the dashboard, press **CTRL+S** or click **Save** in the main toolbar.

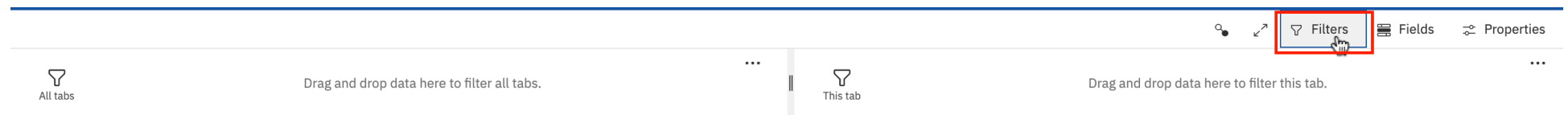
11. Your Panel 5 visualization should look similar to the one below:

Margin by Order Year colored by Product Line

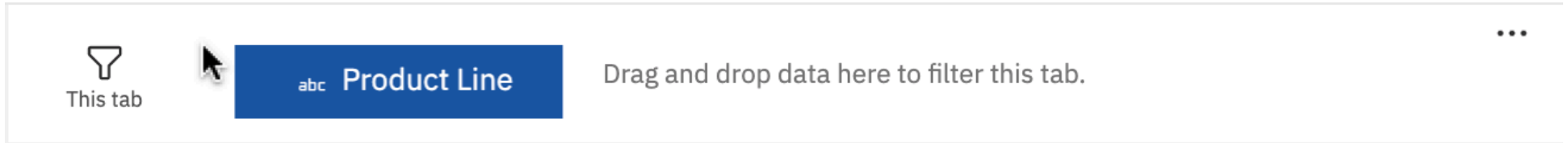


Task E: Filter Data in the Current Tab

1. If required, click **Filters** in the **Dashboard Toolbar** to display the filters pane.



2. From the data source panel, select **Product Line** and drag it to the **This tab** filter panel on the right hand side.



3. Click the **Product Line** filter tab of the **This tab** filter panel. Select **Computers and Home Office**, **Photography**, and **TV and Video Gaming**, then click **Done**.

The screenshot shows a Cognos Analytics dashboard with a 'Product Line' filter dialog box open. The dialog has a title bar 'Product Line' with a close button. Inside, there is a search bar labeled 'Find'. Below it, a list of product lines is shown with checkboxes: 'Computers and Home Office' (checked), 'Kitchen Appliances' (unchecked), 'Photography' (checked), 'Smart Electronics' (unchecked), and 'TV and Video Gaming' (checked). At the bottom of the dialog are 'Clear all' and 'Invert' buttons, and 'OK' and 'Cancel' buttons. A mouse cursor is pointing at the 'OK' button. In the background, a dashboard is visible with a funnel icon labeled 'This tab', a chart titled 'Average Margin', and a legend for 'Product Line' with 'Computers and Home Office' (blue) and 'TV and Video Gaming' (purple). A bar chart for the year 2016 is also visible.

Product Line

Find

- ☒ Computers and Home Office
- ☐ Kitchen Appliances
- ☒ Photography
- ☐ Smart Electronics
- ☒ TV and Video Gaming

Clear all Invert

OK Cancel

Your final dashboard should look similar to the one below. To save the current work in the dashboard, press **CTRL+S** or click **Save** in the main toolbar.



Feel free to change the appearance and layout of the dashboard you have just created.

Congratulations! You have completed this lab.

✓ **Completed**